Dashboards vs notebooks: Choosing the right tool for the job

Criteria	Dashboards	Notebooks
Primary use	Monitoring KPIs, reporting, executive summaries	Data exploration, analysis, storytelling
Target audience	Business users, decision-makers	Data scientists, analysts, researchers
User interaction	Interactive filters, drill-downs (GUI-based)	Interactive via code (Python, R, JS)
Coding required	Minimal or none	Yes, code-based
Visual customisation	Limited to tool's UI capabilities	Highly customisable using code and libraries
Real-time data	Easily supports real-time updates and live data sources	Requires manual or script-based refresh
Shareability	Easily shared via web or embedded in portals	Shared as files (e.g., *.ipynb*) or converted to HTML/ PDF
Tool examples	Apache Superset, Metabase, Grafana, Power BI	Jupyter, Google Colab, Observable
Best suited for	Summarising and presenting insights	Exploring and generating insights
Output format	Visual dashboards, web-based interfaces	Notebook files with narrative, code, and output

Tools of the trade

Here are the best open source tools that are shaping the future of data visualisation and storytelling—from notebooks to dashboards, and everything in between.

Jupyter Notebooks: Still the data scientist's best friend

Today, Jupyter Notebooks is the leading development environment for data science. Within a web browser, users can record live coding, visualisations, and narrative text and equations in one document. This flexible format is useful for either collaborative and reproducible analysis working with data exploration or for data communication.

Historically, Jupyter began with support for Python, but now supports R, Julia and SQL, allowing teams across numerous disciplines to use it. It is simple to conduct data analyses, model development, and data visualisation with library dependencies for pandas, seaborn and Matplotlib within the Jupyter Notebook environment.

What is compelling about Jupyter Notebooks is its ability to log the reasoning behind code execution and thinking in the same document. Analysts can document their exploratory analysis thoughts, the reasoning behind their assumptions, and the graphical representation of discovered patterns in public documentation for other team members to follow.

The functionality of Jupyter Notebooks can be expanded with JupyterLab, Voila and nbconvert. Notebooks can be converted into applications, published as reports and dashboards, etc.

Apache Superset: The open source BI alternative

Superset is an open source business intelligence (BI) application that allows users to build interactive dashboards. It excels at providing enterprise-grade features that do not require any licensing fees.

Superset offers two pathways to analyse data. Users can either drag-and-drop objects into their visualisations or they can use the SQL Lab for writing SQL queries. Superset can connect to dozens of databases, such as PostgreSQL, MySQL, Snowflake, and others, including role-based access.

There are many reasons Superset is gaining traction, one of them being that it is able to generate dynamic dashboards, which can serve live data that refreshes automatically and is aesthetically pleasing. Superset can produce a myriad of chart types, i.e., line graphs, pie charts, heatmaps, treemaps, and time-series analysis. Users can utilise filters on dashboards, drill into data, and link charts together.

Metabase: Simple yet powerful dashboarding

Metabase is an open source BI tool designed for simplicity and speed. It is designed so that users, specifically without great technical skill, can ask questions of their data and receive meaningful answers in beautiful visualisations. The display is intuitive, allowing users to either write SQL or utilise a graphical query builder to search databases, filter tables, and create reports.

Metabase connects to multiple data sources such as PostgreSQL, MySQL, MongoDB, BigQuery, and so on. After being connected, it provides instant visualisations with little configuration—ideal for thought-provoking insights or light dashboarding.